

Abhinita Sanabada

M.S. Data Science Candidate, San José State University | Expected May. 2027

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Seeking Summer/Fall 2026 internships in AI/ML systems, GenAI applications, or software engineering.

SUMMARY

M.S. Data Science candidate focused on AI application engineering and ML/data systems, with 10 years of prior software engineering experience across backend, data, and production systems. Brings hands-on work in healthcare retrieval and semantic-search pipelines, speech/NLP modeling and agentic AI systems, backed by production experience across APIs, distributed pipelines, event-driven systems, and observability.

SKILLS

- **LLM / GenAI / Agents:** retrieval and semantic-search pipelines, agentic AI systems, multi-step agent workflows, tool use and function calling, Model Context Protocol (MCP) tool/server/client integration, human-in-the-loop review, LangChain/LangGraph
 - **AI / ML Engineering:** scikit-learn, NLP, computer vision, CNNs, BiLSTM, attention, regression, classification, model evaluation, metrics/error analysis, class imbalance handling
 - **Data / Distributed Systems:** Spark, PySpark, Kafka, HDFS, MapReduce, Airflow, batch/distributed data processing, event-driven AI/data pipelines
 - **Backend / APIs:** Python, Java, Spring Boot/MVC, REST APIs, FastAPI, GraphQL, Node.js/Express
 - **Cloud / DevOps / Observability:** AWS, Azure, Docker, Kubernetes/Kyma, Jenkins, Linux/Unix, AutoSys, Splunk, ServiceNow
 - **Databases / Caching:** MongoDB, Oracle, MySQL, PostgreSQL, Redis
 - **AI Developer Tools / Testing / Analytics:** Cursor, Claude Code, Codex, JUnit, Selenium, k6, Power BI, Tableau
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AI / ML SYSTEMS PROJECTS

Agentic Recruiting Workflow Simulation for LinkedIn-style Workflows

[link](#)

- Built an **agentic AI workflow system** for multi-step recruiting workflows, including document parsing, candidate-job matching, shortlist generation, outreach drafting, and **human-in-the-loop** review.
- Designed **Kafka-based AI request/result pipelines** with **WebSocket** progress updates, event logging, failure handling, and idempotent consumers to support reliable multi-step workflow execution.
- Implemented the distributed application using **React/Next.js, Node.js/Express, FastAPI, Kafka, MySQL, MongoDB, Redis, Docker**, and **AWS**, with benchmarking on a seeded **10K-record** dataset.

Healthcare Affordability Semantic Search over CMS Pricing Data

[link](#)

- Built an end-to-end **semantic-search and retrieval pipeline** over ~4GB of CMS pricing data (~13.6M rows), using **HDFS, MapReduce, Spark**, and procedure-level data standardization.
- Implemented **MiniLM SentenceTransformer embeddings** to support natural-language procedure search and retrieve relevant procedures with estimated cash prices from ~47K standardized records.
- Added exploratory analysis and **Spark-based regression modeling** to support pricing estimation, affordability comparison, and usability of large public healthcare datasets.

Speech Emotion Recognition with Dialogue Context

[link](#)

- Built a lightweight **speech emotion-recognition model** using **Log-Mel spectrogram CNN features, MFCC-based temporal features, dialogue-context encoding, BiLSTM**, and **attention** to model emotional transitions across conversations.
 - Trained and evaluated on **IEMOCAP** and **MELD** with **focal loss, label smoothing, data augmentation**, and speaker normalization to handle class imbalance and improve minority-class robustness.
 - Achieved **55.9% accuracy** and **52.4% Macro-F1** on IEMOCAP, outperforming MFCC+MLP and transformer-from-scratch baselines while targeting real-time inference.
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WORK EXPERIENCE

Wells Fargo

Nov. 2021 – Jan. 2024

Senior Software Engineer

- Built and owned **Java/Spring backend services, REST APIs, and SQL-backed processing workflows** for the FCM platform, supporting trade capture, clearing/allocations, positions, margin, and risk workflows with improved latency, correctness, auditability, and structured logging.
- Automated and standardized **EOD margin/risk/positions** processing using **AutoSys** job chains on Linux/Unix (dependencies, retries, alerts, runbooks), reducing manual intervention and improving run consistency for risk-sensitive back-office cycles.
- Built operational tooling over runbooks, **Splunk, ServiceNow**, and AutoSys metadata to streamline issue triage,

monitoring, support workflows, and on-call handoffs across production systems.

- Optimized **SQL stored procedures** supporting core processing modules, delivering ~**30% performance improvement** and stronger edge-case correctness through query tuning and validation.

SAP Labs

Apr. 2019 – Oct. 2021

Software Development Engineer 2

- Delivered an automotive orchestration solution deployed on **AWS HANA Cloud**, automating engineering tasks and shift scheduling; built maintainable services and API workflows for product-scale usage.
- Led a **Kyma (Kubernetes)** proof-of-concept to enable cloud-native extensibility; containerized services with **Docker** and improved diagnostics, deployment visibility, and operational stability.
- Owned post-release reliability by monitoring system health, investigating incidents in **Splunk**, managing issues in **ServiceNow**, and shipping hotfixes while supporting secure coding practices including **authN/authZ, encryption, and secure storage**.
- Improved logging, telemetry, and operational documentation for common failure modes, enabling faster triage and more reliable support across multiple environments.

JPMorgan Chase & Co.

Jan. 2017 – Apr. 2019

Software Engineer

- Built backend services and web applications for **asset / wealth management** workflows, delivering Spring Boot REST endpoints and service integrations under Agile delivery.
- Designed persistence and data-access layers across **Oracle** and **MongoDB**, including synchronization and replication-oriented patterns to keep client and position records consistent across systems and support reporting use cases.
- Implemented validation, error handling, and **audit-friendly logging** for sensitive financial transactions with **Kafka** integration; improved release stability through CI-backed testing and controlled deployments.
- Partnered with QA and operations teams to triage production issues, analyze logs, and ship fixes; improved observability and reduced time-to-diagnose by standardizing logs and operational runbooks.

Capgemini

Apr. 2014 – Aug. 2016

Software Engineer

- Built reporting and data solutions for banking clients using Core Java and Oracle; developed **Power BI / Tableau** reports and complex SQL (joins, CTEs, stored procedures) for regulatory and management reporting.
- Improved report reliability and performance by tuning SQL, handling data edge-cases with **PySpark**, and collaborating with stakeholders to translate requirements into production-ready deliverables.

CERTIFICATIONS & RECOGNITION

- Completed Andrew Ng's ML Engineering certificate covering end-to-end ML system design, training, evaluation, and deployment fundamentals.
- Received employee appreciation recognition for ownership and delivery in production, risk-sensitive engineering environments.

EDUCATION

San José State University, San Jose, CA

Jan. 2025 – May. 2027 (expected)

M.S. Data Science — Relevant Coursework: GenAI, ML, Distributed Systems, Cloud Computing, Database Systems, Algorithms

Biju Pattnaik University of Technology

B.Tech. in Electronics and Communication Engineering